

Question Number	Answer	Additional Guidance	Mark
1(a)	A description that includes six of the following points: - description of dependent variable as number of eggs hatched {per unit time/stated time} (1) - use (5) {different/stated} concentrations of copper ions (1) {count brine shrimps hatched/swimming} at {same/stated interval(s)/record time for first hatching} (1) - identify one variable (1) - Identify a second variable (1) - repeats to calculate {mean / SDs} (1) - further method detail (1)	Accept time taken to hatch / time for a fixed/stated number of eggs to hatch Accept a concentration without copper as 0% Accept time for 100% hatching Temperature/salt concentration/pH/species of shrimp/age of eggs/number of eggs/light intensity Accept average e.g. delivery of eggs/making concentrations of copper ions/shining light for counting	6

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1(b)	No ethical concern because: shrimps {are not vertebrates / cannot feel pain / are not harmed (in this experiment) / are plentiful so no risk of extinction}	Candidates will express this in several different ways Accept little pain Ignore simple nervous system/feelings/stress/return to habitat	1

Question Number	Answer	Additional Guidance	Mark
1(c)	A description that includes the following points: - copper ions interfere with {secondary structure / ionic bonding / R groups} (1) - alters {tertiary / 3D} structure of enzyme (1) - active site shape changed (1) {reduces / stops} substrate {fitting/binding} (1)	Ignore refs to pH Accept enzyme denatures Accept copper ions bind to the active site Accept fewer / less or no E-S complexes formed	3

(Total for question 1 = 10 marks)

